

Brussels, 19 May 2026

COST 043/26

DECISION

Subject: Memorandum of Understanding for the implementation of the COST Action “NutriSENSE: A European Network on Bioactives for Personalized Nutrition and Gut Health” (NutriSENSE) CA25129

The COST Member Countries will find attached the Memorandum of Understanding for the COST Action NutriSENSE: A European Network on Bioactives for Personalized Nutrition and Gut Health approved by the Committee of Senior Officials at its 227th meeting held in Brussels on 19 May 2026.

MEMORANDUM OF UNDERSTANDING

For the implementation of a COST Action designated as

COST Action CA25129

**NUTRISENSE: A EUROPEAN NETWORK ON BIOACTIVES FOR PERSONALIZED NUTRITION AND
GUT HEALTH (NutriSENSE)**

The COST Members through the present Memorandum of Understanding (MoU) wish to undertake joint activities of mutual interest and declare their common intention to participate in the COST Action, referred to above and described in the Technical Annex of this MoU.

The Action will be carried out in accordance with the set of COST Implementation Rules approved by the Committee of Senior Officials (CSO), or any document amending or replacing them.

The main aim and objective of the Action is to harmonize and advance European research on bioactive–gut microbiota–health interactions to support sustainable, evidence-based personalized nutrition strategies for preventing non-communicable diseases through interdisciplinary collaboration, shared methodologies, and integrated approaches in food science, nutrition, and sustainability. This will be achieved through the specific objectives detailed in the Technical Annex.

The present MoU enters into force on the date of the approval of the COST Action by the CSO.

OVERVIEW

Summary

NutriSENSE is driven by the conviction that personalised nutrition can address Europe's pressing health and sustainability challenges. Current research on diet, bioactives, and the microbiome is fragmented, limiting reproducibility and translation into policy and innovation. NutriSENSE will overcome these barriers by creating an inclusive, interdisciplinary network that establishes Europe as a leader in personalised nutrition. Its mission is to provide the scientific and societal foundation for health-effective, sustainable, and accessible personalised strategies, contributing to healthier aging and a reduced burden of diet-related non-communicable diseases.

Anchored in the COST mission, NutriSENSE will harmonise methodologies, build interoperable databases, and apply AI tools to generate actionable recommendations. By engaging scientists, policymakers, industry, NGOs, and citizens, it will ensure that knowledge leads to tangible societal benefits.

Inclusiveness is central. A Membership Expansion Team will foster broad participation, with the goal of engaging at least 40 COST countries by the end of the first grant period. Special emphasis will be placed on Inclusiveness Target Countries (ITCs), ensuring access to funding, infrastructures, and leadership roles. Gender equity will be promoted through a Gender Balance Coordinator and Action Plan, while Young Researchers and Innovators (YRIs) will gain visibility through a dedicated Think Tank, mentoring schemes, and leadership opportunities. Policy relevance is strong, with contributions to the European Green Deal, Farm to Fork Strategy, EU4Health, and the UN Sustainable Development Goals. By its conclusion, NutriSENSE will deliver harmonised guidelines, AI models, databases, and toolkits, leaving a durable, inclusive network that continues beyond the Action's lifetime.

Areas of Expertise Relevant for the Action • Agricultural biotechnology: Biotechnology (non-medical) • Agricultural biotechnology: Fermentation	Keywords • Personalised Nutrition • Bioactives–Microbiome Interactions • Healthier Aging • Sustainable Food Systems • AI-Driven Integration
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Specific Objectives

To achieve the main objective described in this MoU, the following specific objectives shall be accomplished:

Research Coordination

- Harmonize definitions and methodologies for studying bioactive–microbiota interactions, including carotenoids, phenolics, organosulfur compounds, phytosterols, and betalains, across European research groups.
- Coordinate experimental protocols and data collection to enable comparable studies across different populations, dietary patterns, and geographic regions.
- Develop integrated databases and comparative analyses that combine bioactive composition, dietary intake, microbiome profiles, and health outcomes.
- Foster interdisciplinary collaboration to link FS&T (production, formulation, LCA) with nutritional and microbiome research, identifying knowledge gaps requiring international coordination.

- Promote collaborative pilot studies and proof-of-concept trials to test bioactive-rich food formulations for personalized nutrition applications.

Capacity Building

- Facilitate knowledge exchange and training through workshops, webinars, and short-term scientific missions (STSMs), connecting researchers from academia, SMEs, and industry.
- Foster interdisciplinary integration by combining knowledge from nutrition, microbiology, food science, computational biology, biotechnology, and sustainability to achieve a holistic understanding of diet–microbiome–health interactions.
- Engage diverse stakeholders, including professionals (food technologists, nutritionists, dietitians), policymakers, regulatory agencies, NGOs, and SMEs, to ensure research outputs are aligned with societal needs and industrial feasibility.
- Support inclusiveness and participation from countries with lower research capacity (ITCs), as well as Near Neighbour Countries (NNCs) and International Partner Countries (IPCs), ensuring equitable access to expertise, networks, and resources.
- Disseminate practical guidelines and open-access resources for researchers and industry on the use of bioactive compounds in personalised nutrition, fostering long-term European leadership in the field.

TECHNICAL ANNEX

1. EXCELLENCE IN S&T AND NETWORKING

A. Main challenge

Background and Rationale

Non-communicable diseases (NCDs) such as obesity, type 2 diabetes, cardiovascular disease, and certain cancers account for the majority of morbidity and mortality in Europe. Dietary strategies are central to prevention and management. Plant- and food-derived **bioactive compounds** - including *carotenoids*, *phenolics*, *organosulfur compounds*, *phytosterols*, and *betalains* - have demonstrated potential to reduce disease risk through antioxidant, anti-inflammatory, lipid-regulating, and microbiota-modulating mechanisms. However, bioactive–health relationships are highly variable among individuals, largely due to differences in gut microbiota composition, genetics, dietary patterns, and lifestyle factors. Current research on these interactions is fragmented, with limited harmonisation in food science and technology (FS&T) methodologies, life cycle assessment (LCA) considerations for sustainable production, and translation into personalised nutrition strategies. This fragmentation hinders the development of functional foods and dietary recommendations that are tailored to both health and sustainability.

Core Problem

Despite growing interest in gut microbiota and nutrition, **there is no pan-European platform** coordinating research on **how diverse bioactive compounds interact with the individual microbiome** to influence health outcomes, nor integrating FS&T, sustainable production (including LCA), and nutrition science in a unified framework. Data, methodologies, and expertise remain siloed across disciplines and countries, leading to duplication of effort, limited statistical power, and low reproducibility.

Main Goal, Main Challenge

NutriSENSE aims to establish the first European network dedicated to harmonising and advancing research on bioactive–microbiota–health interactions for personalised nutrition, spanning the full value chain from **sustainable bioactive sourcing and production to evidence-based dietary applications**. This network will connect *agronomists, analytical chemists, nutritionists, dietitians, bioinformaticians, molecular biologists, microbiologists, food technologists, experts in environmental and techno-economical assessments, and SMEs* to develop shared resources, validated protocols, and an open-access knowledge hub.

The main challenge addressed by NutriSENSE is the fragmented and uncoordinated research landscape on how diverse dietary bioactive compounds - including *carotenoids, phenolics, organosulfur compounds, phytosterols, and betalains* - interact with the **human gut microbiota** to influence health outcomes. Although individual studies have shown promising links between these bioactives and the prevention of NCDs, variability in human responses is high, driven by differences in microbiome composition, genetics, and dietary patterns. Currently, **Europe lacks a unified platform to:**

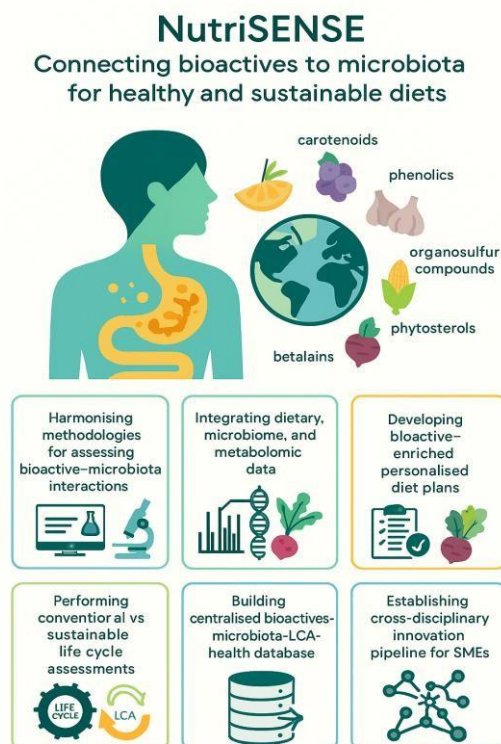
- Harmonise methodologies for bioactive–microbiota research across food science, nutrition, and microbiology;
- Integrate LCA and sustainable production into the bioactive research pipeline;
- Translate findings into personalised nutrition strategies and gut-health-targeted food innovations;
- Facilitate interdisciplinary collaboration and data sharing across borders.

Without such a **coordinated effort**, the potential health, economic, and environmental benefits of bioactives in personalised nutrition will remain underexploited.

Relevance and Timeliness of NutriSENSE

The NutriSENSE Action is highly relevant and timely for several converging reasons: i) **escalating burden of NCDs**: in Europe, obesity, diabetes, cardiovascular diseases, and gastrointestinal disorders continue to rise, placing heavy strain on healthcare systems. There is growing recognition that **dietary strategies targeting gut health** can play a key role in prevention and management. However, current nutritional guidelines rarely consider **individual variability in bioactive responses** mediated by the gut microbiota; ii)

scientific opportunity – gut microbiome as a game changer: over the last decade, the microbiome has been identified as a critical mediator of dietary effects. At the same time, advances in omics, bioinformatics, and systems biology now allow unprecedented insight into **diet–microbiota–host interactions**. The scientific community is thus uniquely positioned to build **harmonised, large-scale, cross-European studies** on bioactives and gut health; iii) **fragmentation of research efforts**: while major progress has been made in understanding individual bioactive compounds, food processing, or sustainability, these fields remain **siloed**. No pan-European network currently integrates **bioactive compound research, personalised nutrition, FS&T, and sustainability (LCA)**. Indeed, there is a high level of specialization among the groups working on bioactives, but it is very uncommon for their studies to encompass more than one or two groups. Cross-talk between groups is essential to consider the whole spectrum of bioactives and gain complete and meaningful insight into their beneficial actions. Without such integration, research outputs remain limited in scope and difficult to translate into practical applications; iv) **societal and economic urgency**: consumers are increasingly demanding **functional foods** that promote gut health and well-being, while industry seeks sustainable solutions for bioactive production (e.g., valorisation of by-products). SMEs, which dominate the European food sector, often lack access to cutting-edge research and collaborative platforms. **NutriSENSE provides a mechanism to bridge academia, industry, and society**; v) **policy alignment and European leadership**: the Action aligns with **EU Green Deal, Farm to Fork Strategy, and Horizon Europe Cluster 6 priorities** on sustainable food systems, nutrition, and health. By building on Europe's strong research base and connecting underrepresented countries, NutriSENSE will strengthen **European leadership in personalised nutrition and sustainable food innovation**.



The timeliness of NutriSENSE is clear: scientific tools such as omics, artificial intelligence, and microbiome profiling are now sufficiently advanced to enable large-scale integration of nutrition and health data. At the same time, public and industry demand for personalised and sustainable nutrition is growing rapidly, while policymakers are calling for systemic approaches that link food, health, and the environment. Without coordinated action, Europe risks ceding leadership in this rapidly evolving field to regions such as North America and Asia, where microbiome-based personalised nutrition and food innovation are advancing quickly. **NutriSENSE is therefore both relevant and urgent**, offering the collaborative platform needed to coordinate fragmented research, accelerate knowledge transfer, and translate scientific progress into tangible societal benefits.

Why Europe: Europe is uniquely positioned to lead this effort thanks to its exceptional diversity in dietary patterns, food traditions, and gut microbiome profiles, which together provide a natural laboratory for **studying bioactive–microbiota–health interactions**. Its expertise in food science, nutrition, microbiology, and sustainability is strong but scattered, and the SME-driven agri-food sector lacks structured access to such knowledge. **COST Actions offer the ideal framework to bridge these gaps by fostering collaboration across all of Europe, including ITCs, ensuring wider participation and equitable capacity building**. By mobilising Europe's diverse resources and cultural assets, **NutriSENSE will generate synergies unattainable by individual countries or bilateral collaborations**, thereby reinforcing Europe's leadership in personalised, sustainable nutrition.

B. Objectives

Research Coordination Objectives (RCOs)

NutriSENSE will *strengthen research coordination by establishing validated and standardised protocols that allow reproducibility of experiments across laboratories and countries, ensuring robust and comparable results*. By leveraging Europe's exceptional diversity of dietary patterns and microbiome profiles, the Action will coordinate cross-national comparative studies that no single research group could achieve alone. A key objective is to guarantee data interoperability by adopting **FAIR principles**, thereby making datasets readily usable for advanced computational analysis and AI-based modelling. NutriSENSE will also bridge experimental research with computational pipelines by coordinating workflows that connect wet-lab discoveries with systems biology and modelling approaches. To secure a long-lasting impact, the Action will produce reference resources, including **the first database on food bioactives**, a compendium of harmonized protocols, white papers on methodological standards, and a shared knowledge hub, to guide future research across Europe and beyond. In this way, NutriSENSE will move the field from fragmented and isolated studies toward a unified, collaborative, and internationally recognised research framework.

NutriSENSE RCOs aim to harmonise methods, align data collection, and integrate interdisciplinary research, creating a common framework to accelerate collaborative studies on bioactives–microbiota–health interactions across Europe. The RCOs include:

RCO1. Harmonize definitions and methodologies for studying bioactive–microbiota interactions, including carotenoids, phenolics, organosulfur compounds, phytosterols, and betalains, across European research groups.

RCO2. Coordinate experimental protocols and data collection to enable comparable studies across different populations, dietary patterns, and geographic regions.

RCO3. Develop integrated databases and comparative analyses that combine bioactive composition, dietary intake, microbiome profiles, and health outcomes.

RCO4. Foster interdisciplinary collaboration to link FS&T (production, formulation, LCA) with nutritional and microbiome research, identifying knowledge gaps requiring international coordination.

RCO5. Promote collaborative pilot studies and proof-of-concept trials to test bioactive-rich food formulations for personalized nutrition applications.

Capacity Building Objectives (CBOs)

NutriSENSE will further strengthen European capacity by investing in the next generation of researchers and innovators. YRIs will be systematically included in leadership roles, mentoring schemes, and Training Schools, ensuring that the network builds long-term expertise across Europe. *Cross-sectoral mobility will be fostered by enabling exchanges not only between academic institutions but also with SMEs, large industry, and policy organisations, equipping participants with a rare blend of scientific, entrepreneurial, and regulatory perspectives*. Beyond the scientific community, NutriSENSE will contribute to **societal capacity** by engaging citizens directly through outreach activities that enhance literacy on personalised and sustainable nutrition, thereby increasing trust and uptake of innovations. The Action will also address the growing demand for digital expertise by providing hands-on training in AI-driven analytics, data stewardship, and FAIR data principles, preparing researchers and industry partners to fully exploit the wealth of nutrition and microbiome data being generated. Finally, *NutriSENSE will ensure that the network's outcomes outlive the Action itself by creating durable structures such as an open-access knowledge hub, long-term stakeholder alliances, and a roadmap for future funding opportunities*. In this way, the Action will leave behind not just scientific knowledge but also a resilient ecosystem of people, skills, and tools to sustain European leadership in personalised and sustainable nutrition.

The CBOs of NutriSENSE focus on strengthening skills, fostering knowledge exchange, and connecting diverse stakeholders across disciplines and countries. The Action will enable collaboration among researchers, industry, policymakers, and SMEs, ensuring inclusive participation and the effective translation of scientific insights into practical applications for personalised and sustainable nutrition. Pointedly, this will be achieved through:

CBO1. Facilitate knowledge exchange and training through workshops, webinars, and short-term scientific missions (STSMs), connecting researchers from academia, SMEs, and industry.

CBO2. Foster interdisciplinary integration by combining knowledge from nutrition, microbiology, food science, computational biology, biotechnology, and sustainability to achieve a holistic understanding of diet–microbiome–health interactions.

CBO3. Engage diverse stakeholders, including professionals (food technologists, nutritionists, dietitians), policymakers, regulatory agencies, NGOs, and SMEs, to ensure research outputs are aligned with societal needs and industrial feasibility.

CBO4. Support inclusiveness and participation from countries with lower research capacity (ITCs), as well as Near Neighbour Countries (NNCs) and International Partner Countries (IPCs), ensuring equitable access to expertise, networks, and resources.

CBO5. Disseminate practical guidelines and open-access resources for researchers and industry on the use of bioactive compounds in personalised nutrition, fostering long-term European leadership in the field.

The objectives of NutriSENSE are directly aligned with the main challenge of fragmented research on bioactives, microbiota, and personalised nutrition. They are relevant because they establish a coordinated European framework for harmonising methodologies, sharing data, and integrating knowledge across diverse populations, dietary patterns, and food cultures. The objectives are concrete and actionable within the 4-year COST Action lifetime, as they rely on specific measures such as workshops, training events, STSMs, pilot studies, and the development of open-access databases. They are achievable because they leverage existing expertise and networks across nutrition, microbiology, food science, computational biology, and sustainability, while engaging YRIs, SMEs, leading companies, and stakeholders from ITCs, NNCs, and IPCs. In addition, the objectives are ambitious: by creating a pan-European, interdisciplinary platform, NutriSENSE aims to overcome fragmentation, generate comparable and high-quality data, and accelerate the translation of scientific insights into personalised and sustainable nutrition solutions, positioning Europe as a global leader in this emerging field.

C. State-of-the-art

State-of-the-art

Over the past decades, research has demonstrated that food-derived bioactive compounds - including **carotenoids, phenolics, organosulfur compounds, phytosterols, and betalains** - can provide health-promoting effects through antioxidant, anti-inflammatory, metabolic, and microbiota-modulating activities. Several European initiatives, such as **BIOCLAIMS** and **Phenol-Explorer**, have advanced knowledge and data availability (Goldansaz et al. 2017). COST Actions, including **EUROCAROTEN (CA15136)** and **POSITIVE (FA1403)**, have significantly contributed to understanding specific bioactives and inter-individual variability in response to their consumption. In parallel, the role of the **gut microbiota** has emerged as a key determinant of variability in bioactive efficacy. Gut microbes can metabolise polyphenols into bioactive metabolites with distinct biological activity, while carotenoids, organosulfur compounds, and phytosterols may themselves shape microbial community structure. However, the evidence remains fragmented, with research conducted on narrow compound classes, isolated populations, or specific health outcomes. **There is no pan-European effort coordinating comprehensive bioactives databases, protocols, clinical studies, and data-sharing on bioactive–microbiota interactions in the context of personalised nutrition.** In **FS&T**, progress has been made in developing innovative extraction and stabilisation processes for bioactives, including supercritical CO₂, ultrasound, and enzymatic methods (Barjoveanu et al. 2020; Nemes et al. 2025; Zhang & Wu, 2025). These techniques improve yield and preserve compound integrity, but research rarely considers downstream effects on bioactive bioavailability, microbiota interactions, and health outcomes. Furthermore, **methodological heterogeneity** across laboratories reduces reproducibility and hinders data integration. On the **sustainability side**, various COST Actions (e.g., **FoodWaStop**, **ENFiber**) and EU projects have addressed waste valorisation and circular bioeconomy approaches. Yet, **technoeconomic evaluation** or **LCA** of bioactive production is seldom connected with nutritional and health research. The gap between **sustainability metrics** and **dietary impact studies** limits the potential to design functional foods that are both **health-promoting and environmentally responsible**.

Still, with all the efforts made, some **key knowledge gaps remain**:

1. Lack of harmonised, cross-country protocols for studying bioactive–microbiota interactions.
2. Fragmentation between FS&T, agrifood production, nutrition, systems biology, microbiology, bioinformatics, clinical studies, and sustainability communities.
3. No **pan-European database** linking bioactive intake, microbiota profiles, health outcomes, and LCA data.

4. Limited translation into **personalised nutrition guidelines** and **functional food innovation**.

Progress Beyond the State of the Art

NutriSENSE will address the gaps by establishing the first **pan-European network** dedicated to **systematically addressing the multi-compound diet–microbiome–personalised nutrition nexus with an integrated FS&T and LCA perspective**. NutriSENSE will go beyond the state of the art by:

- **Covering a broad spectrum of bioactives** - *carotenoids, phenolics, organosulfur compounds, phytosterols, and betalains* - to capture a wide range of health effects and microbiota interactions.
- **Harmonising protocols** for in vitro, in vivo, and human studies of bioactive–microbiota interactions, ensuring methodological consistency and reproducibility across laboratories.
- **Integrating FS&T and nutrition science**, linking food processing and stabilisation methods to downstream bioavailability, microbiota, and health outcomes.
- **Performing technoeconomic analysis and LCAs** to compare conventional vs. sustainable bioactive production and valorisation from agri-food by-products.
- Developing **guidelines for incorporating bioactives into personalised diet plans**, informed by microbiome science, regional food resources, and sustainability constraints.
- Developing the **first European open-access database** that connects dietary bioactive intake, microbiota profiles, metabolomics, health biomarkers, and environmental performance.
- Establishing a **cross-disciplinary innovation hub** to support SMEs in designing sustainable, gut-health-targeted food products adapted to diverse European dietary patterns.

Through these advances, NutriSENSE will overcome the current fragmentation of research efforts, enable reproducible and translatable findings, and provide the scientific and technological foundation for personalised nutrition strategies that are both **health-effective** and **sustainable**. By fostering evidence-based dietary recommendations tailored to individual needs, the Action will contribute to healthier aging, reduced NCD burden, and improved quality of life across Europe's diverse populations, while also supporting innovation and sustainability within food systems.

Uniqueness of NutriSENSE

NutriSENSE does not duplicate existing efforts but instead complements and expands them by addressing critical gaps at the intersection of bioactives, the gut microbiome, and personalised nutrition. Whereas many initiatives focus narrowly on either dietary components, microbiome science, or sustainability in isolation, NutriSENSE uniquely integrates these dimensions within a coordinated pan-European network. By harmonising methodologies, fostering data interoperability, and engaging stakeholders across academia, industry, and policy, **the Action creates synergies that no single project or bilateral collaboration could achieve**. Its added value lies in bridging fragmented expertise, enabling reproducible and translatable findings, and ensuring that scientific knowledge is effectively channelled into societal and industrial innovation. This positioning ensures NutriSENSE is complementary to ongoing research while filling unmet needs in coordination, knowledge transfer, and policy alignment, thereby reinforcing Europe's leadership in personalised and sustainable nutrition.

D. Rationale for choosing networking to address the main challenge

Rationale for Networking Approach

The **main challenge** - *fragmentation of research on bioactives, microbiota, and personalised nutrition* - **cannot be effectively addressed by isolated research projects alone**. Studies are dispersed across disciplines, countries, and institutions, with diverse methodologies, datasets, and population cohorts. A pan-European network like NutriSENSE provides the **ideal framework** to coordinate these efforts, harmonise methods, and enable knowledge sharing at scale. By connecting researchers, industry partners, SMEs, policymakers, and stakeholders from **European countries (with a focus on ITCs) and other geographical areas**, the Action fosters interdisciplinary collaboration, capacity building, and equitable participation. Networking ensures a fluent and effective interaction, the rapid dissemination of best practices, the integration of fragmented data, and the creation of common tools and standards, thereby accelerating innovation and societal impact in personalized and sustainable nutrition - outcomes that individual research projects alone could not achieve.

Justification for a Pan-European Network

Establishing a pan-European network is the most suitable approach for NutriSENSE, as the main challenge is inherently interdisciplinary and transnational. Scientific expertise, population diversity, dietary patterns, and food cultures vary widely across Europe, making coordinated collaboration

essential to generate robust, comparable data. The network structure enables harmonisation of research methodologies, shared protocols, and integrated analyses across countries, while facilitating knowledge exchange between academia, industry, SMEs, policymakers, and innovators. By involving countries from different regions and levels of research intensity, the network ensures equitable access to training, resources, and opportunities for early-career researchers and innovators. This pan-European approach maximises the impact, reach, and efficiency of the Action, allowing NutriSENSE to achieve its objectives - strengthening European scientific coordination, building capacity, and translating insights into personalised and sustainable nutrition - far beyond what isolated projects could accomplish.

NutriSENSE will be the first large international network encompassing all the main groups of bioactives, from production to health benefits, focusing on the interaction with the microbiota and sustainability aspects. Such an ambitious consortium requires a framework (in terms of governance, geographical scope, and activities funded) perfectly fitted to the COST Actions offer to be implemented and achieve its objectives.

E. Critical mass of the network

NutriSENSE brings together a robust network of leading European experts across nutrition, microbiology, food science, computational biology, biotechnology, and sustainability. The network will include researchers from top universities, research institutes, and technology centers with complementary expertise in bioactives, microbiota analysis, omics technologies, bioinformatics, personalized nutrition, and LCA. The involvement of industry partners, governmental agencies, and non-profit organisations will ensure that scientific discoveries are rapidly translated into practical innovations and market-relevant solutions. This academia–industry collaboration fosters the co-creation of knowledge, accelerates technology transfer, and strengthens the link between fundamental research and real-world application, supporting the development of sustainable and health-promoting food systems across Europe. This diverse expertise ensures the network has the critical mass required to coordinate methodologies, harmonize experimental protocols, and integrate multi-dimensional datasets across European populations and dietary patterns. Where current coverage is limited, NutriSENSE includes a clear and credible expansion strategy: targeted outreach to additional research groups in underrepresented countries, engagement of more SMEs and industry partners working on gut-health-related food products, and recruitment of specialized laboratories with capabilities in carotenoid, phenolic, organosulfur, and betalain analysis. Structured working groups, regular networking meetings, workshops, conferences, training schools, and STSMs will ensure meaningful participation, fostering collaboration and integration of all relevant stakeholders. In addition, *Conference Grants* (supporting YRIs as well as participants from ITCs) will facilitate attendance at international scientific events beyond the Action's own activities, broadening visibility and knowledge exchange. Moreover, *Virtual Mobility Grants* will further enhance collaboration by supporting online interactions, digital exchanges, and virtual teamwork across countries - ensuring continuous knowledge flow, skill development, and inclusiveness even beyond physical mobility opportunities. This approach guarantees that the network's scale and breadth are sufficient to achieve the Action's objectives efficiently and effectively.

2. IMPACT

A. Impact related to objectives

NutriSENSE is designed to deliver realistic and innovative impacts that extend from scientific advancement to tangible socio-economic, environmental, and policy benefits. *In the short term*, the Action will generate immediate scientific impact by harmonising methodologies for studying bioactive–microbiota–health interactions and by fostering interdisciplinary dialogue across nutrition, microbiology, food science, computational biology, and sustainability. Through **COST networking tools**, NutriSENSE will rapidly strengthen research coordination, overcoming fragmentation and enabling robust comparative studies across Europe. *In the medium term*, NutriSENSE will build capacity by training a new generation of researchers and innovators, equipping them with skills in omics, microbiome profiling, artificial intelligence, and LCA. The Action will also create structured engagement channels between academia, SMEs, and larger food companies, accelerating the translation of scientific insights into personalised nutrition solutions and functional food innovations. By mobilising expertise across

countries and disciplines, the network will enable SMEs - who represent the backbone of the European agri-food sector - to access harmonised science and innovation pathways that would otherwise be beyond their reach. *In the long term*, the Action **will contribute to healthier, more sustainable diets, with direct benefits to public health and the circular bioeconomy**. By coordinating fragmented efforts across Europe, *NutriSENSE will provide a credible evidence base for policy initiatives linking health, environment, and food systems* (e.g. Farm-to-Fork Strategy, EU Green Deal, Horizon Europe missions). The inclusive and pan-European nature of the network will ensure that benefits are widely distributed, amplifying societal impact and reinforcing Europe's global leadership in personalised nutrition and food innovation.

Through this staged pathway - short-term coordination, medium-term capacity building, and long-term societal and policy impact - NutriSENSE demonstrates how a COST Action can realistically and innovatively address its main challenge while ensuring sustained European competitiveness in this rapidly evolving field.

NutriSENSE is ambitious yet grounded in achievable outcomes, as it builds on existing expertise and infrastructures while filling clear gaps in coordination. Its **scientific impact** will stem from creating a unified framework to study diet–microbiome–bioactive interactions across Europe's diverse populations, an approach that is both novel and realistic given the maturity of omics and AI-based tools. The **socioeconomic impact** lies in empowering consumers with evidence-based personalised nutrition strategies that support healthier aging and reduce the burden of diet-related chronic diseases and NCDs while promoting sustainable dietary choices - an innovation that addresses both health and environmental priorities simultaneously. **NutriSENSE will result in health-related savings to citizens and national health services and wealth creation for innovative companies**. From a **policy perspective**, the Action provides a timely opportunity to inform European strategies linking food systems, health, and sustainability, offering harmonised scientific evidence where fragmented data currently limits policy action. For the **food industry**, and especially SMEs, the innovative impact will come from access to shared databases, harmonised methodologies, and transnational collaborations that lower barriers to innovation and accelerate product development. Moreover, the inclusion of LCA will result in more sustainability-conscious product developments, with the concomitant environmental benefits. In the **long term**, NutriSENSE will establish Europe as the reference hub for personalised nutrition research and innovation, creating durable structures for knowledge exchange that extend beyond the lifetime of the Action.

*Summary of expected Impacts of the NutriSENSE COST Action: **scientific, social, environmental, and policy (political)** impacts across the **short-, medium-, and long-term horizons***

Timeframe	Scientific Impact	Social / Economic Impact	Environmental Impact	Policy / Political Impact
Short-term (Years 1–2)	• Harmonisation of methodologies for studying bioactive–microbiota–health interactions. • Establishment of interdisciplinary dialogue across nutrition, microbiology, food science, AI, and sustainability. • Strengthened research coordination and comparability across Europe.	• Enhanced collaboration between academia, SMEs, and industry. • Early engagement and training of YRIs • Increased inclusiveness through participation of ITC partners.	• Initial mapping of sustainable dietary patterns and food production practices. • Inclusion of LCA experts in the network to integrate environmental indicators.	• Creation of a unified scientific foundation to inform EU priorities in food, health, and research. • Early contribution to <i>Farm to Fork Strategy</i> and <i>Horizon Europe Missions</i>.

Medium-term (Years 2–3)	• Development of interoperable, open-access databases integrating omics, microbiome, and nutrition data. • Capacity building in AI, bioinformatics, and systems biology. • Promotion of reproducibility and harmonised standards.	• Empowerment of SMEs through shared methodologies and innovation access. • Faster translation of research into personalised nutrition and functional food innovations. • Career growth and cross-border mobility for researchers.	• Application of LCA in food innovation to measure and reduce environmental footprints. • Promotion of sustainable sourcing and production strategies. • Increased awareness of nutrition–environment linkages.	• Evidence-based support for the <i>EU Green Deal</i>, <i>EU4Health</i>, and <i>Europe’s Beating Cancer Plan</i>. • Alignment with national sustainability and food policies. • Enhanced stakeholder participation in science-policy dialogue.
Long-term (Years 3–4 and beyond)	• Establishment of Europe as the global reference hub for personalised nutrition and microbiome research. • Sustained integration of omics, AI, and bioactive research infrastructures. • Continued knowledge exchange beyond the Action’s lifetime.	• Improved public health outcomes through healthier aging and reduced NCD burden. • Health cost savings for citizens and healthcare systems. • Competitive growth for innovative European food companies.	• Mainstream adoption of sustainable, health-promoting diets. • Reduced food waste and carbon emissions through evidence-based dietary transitions. • Long-term contribution to circular bioeconomy goals.	• Long-term support for EU and international frameworks linking health, environment, and food systems. • Evidence-based input to dietary guidelines and sustainability policies. • Reinforcement of Europe’s global leadership in food innovation and health policy.

Objective–Impact Mapping

- **Develop a shared framework and definitions for studying bioactive–microbiota–health interactions:** i) *scientific impact (short-term)*: establishes a common language and reference point across disciplines, reducing duplication and inconsistencies; ii) *policy impact (medium-term)*: provides a harmonised evidence base that policymakers can use to design nutrition and health strategies aligned with EU priorities.
- **Coordinate methodologies for analysing diet, microbiome, and bioactives:** i) *scientific impact (short-term)*: enhances reproducibility and comparability of results across Europe; ii) *industrial impact (medium-term)*: reduces the risk for SMEs investing in personalised nutrition by ensuring robust and standardised testing methods.
- **Create shared databases and platforms for comparative analysis of European dietary patterns and gut microbiota diversity:** i) *scientific impact (short- to medium-term)*: enables cross-border data integration not feasible by individual projects; ii) *societal impact (long-term)*: supports evidence-based dietary guidelines tailored to diverse populations.
- **Foster interdisciplinary collaboration between nutrition, microbiology, computational biology, food science, and sustainability:** i) *scientific impact (short-term)*: breaks silos and accelerates innovation at the intersection of disciplines; ii) *innovative impact (medium- to long-term)*: leads to holistic personalised nutrition strategies that also account for environmental sustainability.
- **Engage SMEs, industry leaders, and other stakeholders in co-developing new approaches to personalised nutrition:** i) *industrial impact (medium-term)*: creates new

business opportunities in food innovation, functional ingredients, and digital health tools; ii) *societal impact (long-term)*: facilitates consumer access to safe, affordable, and sustainable personalised nutrition solutions.

- **Build research capacity through training, mobility, and networking with a focus on YRIs:**
 - i) *capacity-building impact (short-term)*: expands the skills base and ensures knowledge transfer to the next generation; ii) *scientific and societal impact (long-term)*: ensures Europe maintains a strong, inclusive, and competitive workforce in personalised nutrition.

B. Involvement of stakeholders

NutriSENSE will build a ***pan-European network that actively engages a broad range of stakeholders across academia, industry, policy, healthcare, civil society, and international partners*** to maximise impact and ensure inclusiveness.

NutriSENSE Stakeholders and Their Roles

- **Academic and Research Institutions**
 - Represent the scientific core of the network, contributing expertise in **nutrition, microbiology, computational biology, food science, biotechnology, and sustainability**.
 - Lead the **generation of new knowledge**, design of research frameworks, and **harmonisation of methodologies** across Europe.
 - Train the **next generation of researchers and innovators**, fostering interdisciplinary collaboration and excellence.
 - Support open data, reproducibility, and cross-border mobility of scientific expertise.
- **Industry Partners (with a focus on SMEs)**
 - Engage companies from the **agri-food, biotechnology, and digital health** sectors to bridge research and application.
 - Participate in the **co-development and validation of methodologies**, ensuring scientific outcomes are relevant for innovation.
 - Test and implement results in **pilot or pre-commercial settings**, facilitating real-world validation.
 - Contribute to **competitiveness and innovation** while addressing the challenges SMEs face in accessing advanced research and infrastructures.
- **Policy Makers and Regulatory Bodies**
 - Include representatives at **EU, national, and regional levels**
 - Participate in **structured policy dialogues, thematic roundtables, and evidence-sharing events**.
 - Use NutriSENSE outputs to inform **policy frameworks on nutrition, food labeling, health claims, and sustainability**.
 - Help ensure that scientific insights translate into practical and regulatory impacts across Europe.
- **Healthcare Professionals and Nutrition Experts**
 - Serve as key intermediaries between research and end-users.
 - Test **personalised nutrition strategies in real-world healthcare and clinical settings**.
 - Provide essential feedback on **feasibility, safety, efficacy, and consumer adherence**.
 - Help to integrate scientific findings into **public health strategies and dietary recommendations**.
- **Civil Society and Consumer Organisations**
 - Represent **end-users, patients, and citizens**, ensuring inclusiveness and diversity.
 - Contribute perspectives on **ethical, cultural, and societal dimensions** of personalised nutrition.
 - Engage in **citizen science activities** and outreach initiatives to enhance awareness and trust.

- Help design communication strategies that are **transparent, accessible, and culturally sensitive**.
- **International Collaborators (IPCs, NNCs, and Global Networks)**
 - Provide **external expertise, comparative datasets, and complementary infrastructures** from outside the EU.
 - Strengthen **global benchmarking** and increase Europe's visibility and leadership in personalised nutrition.
 - Promote international collaboration, data exchange, and alignment with global research agendas (e.g., WHO, FAO).

Overall approach: *Each stakeholder group is not only a passive participant but an active co-creator of knowledge. Tailored engagement strategies - from scientific workshops and policy roundtables to industry-focused innovation hubs and public outreach - will ensure that their contribution is meaningful and sustained throughout the Action.*

Risks and Mitigation

Engaging such a diverse range of stakeholders also presents challenges which will be addressed by the Action MC. SMEs and start-ups may face limited time and financial resources to participate consistently, while policymakers may be constrained by long regulatory cycles and competing agendas. Consumer organisations may express scepticism regarding the safety, accessibility, or equity of personalised nutrition approaches. To mitigate these risks, **NutriSENSE will adopt flexible engagement formats** such as online workshops and short-term innovation sprints to accommodate SMEs, while producing concise, evidence-based policy briefs tailored to decision-makers' needs. For **consumers** and **civil society**, transparent communication, citizen science initiatives, and inclusive dialogues will ensure that ethical, cultural, and accessibility concerns are addressed from the outset. By anticipating and proactively managing these challenges, the Action will secure robust, balanced, and meaningful stakeholder participation throughout its lifetime.

C. Communication, dissemination and valorisation

Communication

NutriSENSE will implement a **proactive, inclusive, and consistent communication strategy** to raise awareness of its goals, activities, and outcomes across all stakeholder groups. A dedicated Action website, launched within the first six months, will host news, updates, resources, and an interactive forum for partners. Regular newsletters, distributed quarterly, will keep network members, stakeholders, and policymakers informed, while social media channels (Facebook, Instagram, Twitter/X, LinkedIn, YouTube) will provide monthly updates, highlight key achievements, and engage the broader public. Multimedia formats such as infographics, podcasts, virtual newspapers, TV, and radio features will complement these efforts, ensuring accessibility and tailored messaging for different audiences. Citizen engagement will be promoted through public events and science cafés in selected cities, designed to build trust, enhance understanding of personalised nutrition, and increase awareness of bioactives and sustainable diets. Key performance indicators (KPIs): *i) 1 dedicated website, updated monthly; ii) 4 newsletters per year; iii) 12 social media posts per quarter; iv) 1 public engagement event per year.*

Dissemination

The dissemination strategy will focus on **sharing scientific outputs with the research community**. This will include open-access publications in peer-reviewed journals, presentations at international conferences, collaborative white papers, and harmonised datasets made available through interoperable repositories. NutriSENSE will organise 2 training schools and 4 thematic webinars per year, targeting YRIs to build capacity in bioactives–microbiome–nutrition research. Harmonised datasets and protocols will be deposited in open-access repositories to facilitate cross-border collaboration. Policy briefs summarising actionable evidence will be prepared annually and distributed to European and national policymakers. **KPIs:** *i) 8–10 open-access publications during the Action; ii) 1 training schools per year; iii) 2 webinars per year; iv) 1 policy brief annually.*

Valorisation

Valorisation efforts will be directed towards the **agri-food industry, especially SMEs**, which stand to benefit from new methodologies, protocols, and innovations generated by NutriSENSE. Innovation showcases, stakeholder roundtables, and pilot collaborations will serve as bridges between research

and practice, enabling industry partners to explore new opportunities for product development in personalised nutrition and sustainable food systems. Civil society organisations will also be involved to ensure that societal perspectives and ethical considerations are integrated into the valorisation process. **KPIs:** i) 4 stakeholder roundtables over 4 years; ii) 1 innovation showcases per year; iii) 4 pilot collaborations with industry partners.

Exploitation & IPR

NutriSENSE will **promote open science** while **respecting intellectual property rights**. Shared resources and protocols will be openly available, while proprietary innovations emerging from cross-sector collaborations will be supported through IPR guidelines to ensure fair benefit-sharing. SMEs and industry partners will be encouraged to explore licensing opportunities, spin-off ventures, and public-private partnerships, ensuring that Action outputs translate into tangible market and societal innovations. This exploitation framework will maximise the long-term sustainability and impact of NutriSENSE outcomes beyond the lifetime of the Action. **KPIs:** i) open-access sharing of all non-proprietary outputs; ii) IPR agreements in place for all industrial collaborations; iii) at least 2 commercialisation initiatives or licensing agreements by the end of the Action.

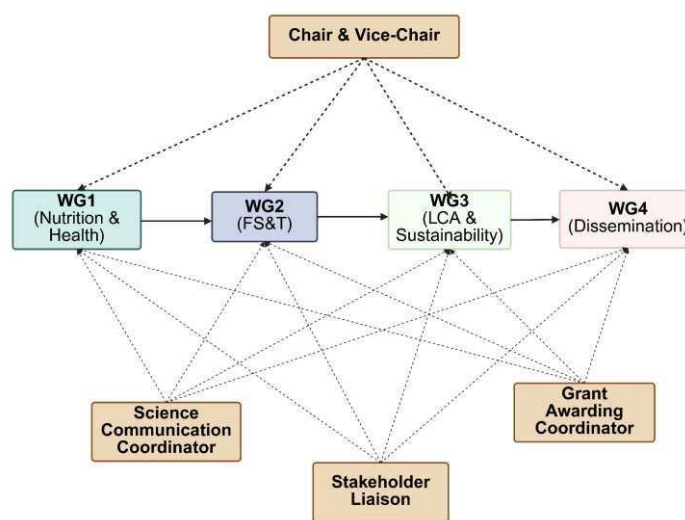
3. IMPLEMENTATION

A. Action Structure

The structure of NutriSENSE is designed to ensure effective coordination, integration of expertise, and delivery of impactful outcomes. The *Action will be organised into four interlinked Working Groups (WGs)*, each addressing a specific dimension of the main challenge while contributing to the overall objectives:

- **WG1 – Bioactive–Microbiome Interactions (Nutrition&Health):** Focuses on mapping and understanding how dietary bioactives (*carotenoids, phenolics, organosulfur compounds, phytosterols, betalains, etc.*) interact with the gut microbiome and influence host health. It will harmonise protocols, share datasets, and provide the biological foundation for personalised nutrition strategies.
- **WG2 – Harmonisation of Methodologies and Standards (FS&T):** Coordinates the development and alignment of experimental designs, analytical pipelines, and protocols across participating labs. This includes omics methods, microbiome profiling, dietary assessment, and in vitro/in vivo testing, ensuring data comparability and reproducibility across Europe.
- **WG3 – Data Integration, AI and Systems Approaches (LCA & Sustainability):** Develops frameworks for integrating multi-omics, clinical, and dietary data, applying advanced computational and AI tools. This WG will build interoperable databases and models to translate complex interactions into actionable personalised nutrition guidelines.
- **WG4 – Stakeholder Engagement, Dissemination and Policy Impact:** Connects researchers with key stakeholders (food industry, SMEs, policymakers, NGOs, and citizens). It will lead to communication, dissemination, and training activities, ensuring that scientific findings are translated into societal, industrial, and policy benefits.

The Action's structure is deliberately designed to maximise efficiency, inclusiveness, and impact. The Management Committee, supported by the **Action Chair, Vice-Chair**, and Working Group Leaders, will oversee coordination, ensure progress toward objectives, and maintain alignment across the four WGs. A **Science Communication Coordinator** and a **Grant Awarding Coordinator** will further support dissemination and mobility activities, ensuring transparency and accessibility for all members. This organisational framework provides clear leadership while remaining flexible enough to adapt to emerging



needs and opportunities, enabling NutriSENSE to operate as a dynamic, collaborative platform. By combining strong governance with specialised expertise in each WG, the Action ensures that knowledge flows seamlessly across disciplines, stakeholders are actively engaged, and outputs are consistently translated into practical and impactful outcomes. Beyond the mandatory COST Action roles (Chair, Vice-Chair, Working Group Leaders, Science Communication Coordinator, and Grant Awarding Coordinator), NutriSENSE will appoint a **Data Management Coordinator** to oversee data sharing, database development, and FAIR (Findable, Accessible, Interoperable, Reusable) data principles. This role is essential to address the main challenge of fragmented and non-standardised datasets. Additionally, a **Stakeholder Liaison Officer** will be designated within WG4 to ensure structured engagement with SMEs, policymakers, and civil society, supporting the Action's capacity-building and impact goals. Together, this structure ensures that NutriSENSE combines strong scientific coordination with proactive stakeholder involvement, effectively tackling fragmentation in the field and advancing personalised and sustainable nutrition across Europe.

Interrelations and Contribution to Objectives

The four WGs of NutriSENSE are designed to operate in strong interconnection, each addressing a specific dimension of the main challenge while mutually reinforcing one another.

WG1 -> *provides the biological foundation for the Action* by generating and harmonising knowledge on how key dietary bioactives interact with the gut microbiome and influence host physiology. Its outputs-datasets, mechanistic insights, and harmonised protocols-are directly taken up by **WG2** -> *which ensures that experimental approaches, analytical pipelines, and dietary assessment methods are aligned across countries*, enabling reproducibility and comparability of findings. The outputs of WG1 and WG2 -> flow into **WG3**, *which develops integrative frameworks to combine multi-omics, dietary, and clinical data*. By applying advanced computational and AI tools, WG3 translates fragmented datasets into predictive models and actionable personalised nutrition strategies, while also incorporating sustainability and LCA aspects to balance health and environmental priorities. **WG4** -> *acts as the transversal hub, connecting scientific advances from the other WGs with external stakeholders*. It ensures that harmonised methodologies (WG2), biological insights (WG1), and data-driven models (WG3) are disseminated to researchers, industry, policymakers, and the wider public. WG4 also fosters capacity building by training YRIs, engaging SMEs, and creating science-policy interfaces that maximise societal and policy impact.

Together, the **WGs contribute to the objectives of NutriSENSE** by:

- Establishing a common scientific and methodological framework across Europe (**WGs 1 & 2**).
- Enabling data integration and systems-level understanding through AI-driven tools (**WG3**).
- Translating scientific advances into societal, industrial, and policy benefits via structured engagement and dissemination (**WG4**).

This interconnected structure ensures that the Action does not operate as fragmented research silos but as a pan-European, synergistic network, capable of delivering the coordination, capacity building, and impact required to advance personalised and sustainable nutrition.

B. Work plan (tasks, activities and timeframe)

The NutriSENSE Action will run for **48 months**, organised into four interlinked WGs. Each WG is responsible for specific tasks, but all activities are aligned to ensure synergies and cross-fertilisation. The plan includes annual milestones and structured networking activities (*workshops, conferences, training schools, STSMs, stakeholder events*), which progressively build toward the achievement of the Action's objectives.

Year 1 – Foundation and Network Consolidation

- **Tasks & Activities:**
 - Establishment of the Management Committee (MC), WG Leaders, and operational guidelines.
 - Kick-off conference (*first annual conference*) to define shared vision and work plan.
 - Mapping of existing expertise, infrastructures, and datasets across participating institutions.
 - WG1: Initial scoping of bioactive-microbiome interactions and identification of priority compounds.
 - WG2: Survey of existing protocols; initiation of harmonisation efforts.

- WG3: Assessment of existing data platforms and computational tools.
- WG4: Launch of Action website, social media, and first newsletter.
- **Milestones:** *i) shared glossary/definitions across WGs; ii) initial database of Action participants' expertise and datasets; iii) communication/dissemination strategy in place.*

Year 2 – Harmonisation and Methodological Alignment

- **Tasks & Activities:**
 - WG1: Comparative analysis of microbiome–bioactive interactions across European cohorts.
 - WG2: Development of harmonised protocols (omics, microbiome profiling, dietary assessment, cooking/cuisine-based FS&T).
 - WG3: Initiation of pilot data integration activities and design of AI models.
 - WG4: Organisation of the first Training School (young researchers, ITCs), the *second annual conference*, and the first policy dialogue workshop.
- **Milestones:** *i) harmonised methodological guidelines validated by Action participants; ii) first stakeholder workshop with SMEs and policymakers; iii) YRI network formally launched.*

Year 3 – Data Integration and Translation

- **Tasks & Activities:**
 - WG1: Refinement of mechanistic insights into diet–microbiome–bioactive interactions.
 - WG2: Dissemination of harmonised standards; cross-lab ring tests.
 - WG3: Development of integrated multi-omics and AI-driven models for personalised nutrition and sustainability.
 - WG4: Training School on data science and communication for YRIs; third annual conference; citizen engagement events (science cafés).
- **Milestones:** *i) prototype of interoperable database established; ii) draft personalised nutrition framework (health + sustainability dimensions); iii) cross-disciplinary workshops linking WGs 1–3 with external stakeholders.*

Year 4 – Consolidation, Dissemination, and Building

- **Tasks & Activities:**
 - WG1: Final synthesis of bioactive–microbiome–health evidence.
 - WG2: Publication of best-practice guidelines and protocols.
 - WG3: Finalisation of AI-based predictive models and sustainability assessments.
 - WG4: Large-scale dissemination conference, targeted policy briefs, and industry roundtables.
 - Preparation of a roadmap for post-Action collaborations and funding opportunities.
- **Milestones:** *i) public release of databases, harmonised protocols, and policy briefs; ii) publication of final reports and scientific white papers; iii) final conference showcasing Action achievements and long-term sustainability strategy.*

Timeframe Summary: *Year 1: Network building, scoping, mapping. Year 2: Harmonisation of methodologies, stakeholder engagement. Year 3: Data integration, translation into models. Year 4: Consolidation, dissemination, sustainability planning.*

C. Deliverables

Deliverables are the tangible outputs of NutriSENSE, ensuring that the Action's objectives translate into concrete, measurable results. They include *scientific reports, critical reviews, roadmaps, harmonised protocols, databases, white papers, policy briefs, and training materials*, each aligned with specific Working Groups. Scheduled across the 4 years, these deliverables provide structured milestones that demonstrate progress, foster knowledge exchange, and maximise the Action's impact on science, society, industry, and policy.

NutriSENSE Deliverables Mapped to Working Groups

Year 1

- **D1.1: Inventory and mapping report** → WG4 (Stakeholder Engagement & Dissemination, with input from all WGs).
- **D1.2: Consensus document on definitions and terminology** → WG1 & WG2 (Nutrition/Health + FS&T).

- **D1.3: Draft protocol compendium** → WG2 (FS&T – harmonisation and methodological standards).

Year 2

- **D2.1: Interim white paper on bioactive–microbiome–health interactions** → WG1 (Nutrition & Health).
- **D2.2: Online database prototype** → WG3 (Data Integration & Sustainability).
- **D2.3: Best practice guidelines for experimental harmonisation** → WG2 (FS&T).

Year 3

- **D3.1: Systems integration framework (omics + AI)** → WG3 (Data & Systems).
- **D3.2: Case study portfolio on personalised nutrition approaches** → WG1 & WG2 (Nutrition/Health + FS&T).
- **D3.3: Policy brief series (first edition)** → WG4 (Dissemination & Policy).

Year 4

- **D4.1: Final white paper – roadmap for Europe** → WG1 & WG4 (Nutrition/Health + Dissemination & Policy).
- **D4.2: Comprehensive open-access database** → WG3 (Data Integration).
- **D4.3: Industry-oriented toolkit for SMEs/food companies** → WG2 & WG4 (FS&T + Dissemination).
- **D4.4: Final policy brief series (second edition)** → WG4.
- **D4.5: Training materials for YRIs** → WG4 (in collaboration with all WGs).
- **D4.6: Dissemination compendium (brochure + digital package)** → WG4.

The deliverables of NutriSENSE are embedded in a clear 4-year plan with defined milestones, ensuring steady progress and measurable outcomes. Their development will actively promote the participation of ITCs, guaranteeing that knowledge and opportunities are shared across Europe. Moreover, gender balance and the meaningful inclusion of YRIs will be integral to each stage of the process, reinforcing equity, diversity, and sustainability in building the next generation of leaders in personalised and sustainable nutrition.

D. Gantt chart

The Gantt chart for NutriSENSE illustrates a **4-year timeline (48 months)** where the four WGs operate in parallel but with coordinated milestones. **WG1 (Nutrition & Health: Bioactive–Microbiome Interactions)** starts immediately (Months 1–6) with protocol harmonisation and dataset mapping. Main milestones include a European database of bioactive–microbiome studies (Month 18) and guidelines for personalised nutrition applications (Month 36). *Deliverables include scientific reports, joint review papers, and a white paper by Month 48.* **WG2 (Food Science & Technology)** runs throughout the Action, beginning with methodology alignment (Months 1–12). Milestones include harmonised lab protocols (Month 12), validated cooking/cuisine models (Month 24), and best-practice recommendations for FS&T (Month 42). *Deliverables include methodological handbook, standardised protocols, and input to standards.* **WG3 (Sustainability: Data Integration & AI)** starts in Year 1 with the design of interoperable data systems and AI pipelines (Months 1–12). Major milestones are the development of a shared data platform (Month 24), predictive AI models (Month 36), and a validated decision-support system for personalised diets (Month 48). *Deliverables include an open-access database, technical diagrams, and software prototypes.* **WG4 (Dissemination, Stakeholders & Policy Impact)** engages with the communication setup from the start (Months 1–6). Milestones include the launch of the website and social media channels (Month 6), first stakeholder workshop (Month 12), annual public events (Years 2–4), and final policy white paper (Month 48). *Deliverables include newsletters, stakeholder reports, training school materials, policy briefs, and outreach content.* Across all WGs, **cross-cutting milestones** include mid-term evaluation (Month 24), training schools for YRIs (Yearly), and final Action conference (Month 48). The chart highlights inclusiveness measures, ensuring active participation from ITCs, gender balance, and YRIs' engagement at every stage.

NutriSENSE Gantt Chart

WG	Task/Activity	Milestone (Month)	Deliverable (Month)	Y1 S1 (M1–6)	Y1 S2 (M7–12)	Y2 S1 (M13–18)	Y2 S2 (M19–24)	Y3 S1 (M25–30)	Y3 S2 (M31–36)	Y4 S1 (M37–42)	Y4 S2 (M43–48)
Cross-cutting	Kick-off, MC & WG setup, guidelines	MS1.1: Shared glossary & definitions (M6)	–	•							
	Mid-term review & adjustments	MS2.1: Interim evaluation (M24)	–				•				
	Inclusiveness (ITC, gender, YRIs)	–	Annual inclusion reports (M12, 24, 36, 48)		•		•				•
	Final conference & sustainability roadmap	MS4.1: Legacy roadmap (M48)	D4.6 Dissemination compendium (M48)								•
WG1 – Nutrition & Health	Scoping bioactive–microbiome interactions	MS1.2: Priority compounds identified (M12)	D1.1 Mapping report (M12)		•						
	Comparative cohort analysis	MS2.2: Interim findings (M24)	D2.1 Interim white paper (M24)				•				
	Mechanistic refinement	MS3.1: Mechanistic map (M36)	D3.2 Case study portfolio (M36)					•			
	Final synthesis	MS4.2: Evidence synthesis (M48)	D4.1 Final white paper (M48)								•
WG2 – FS&T	Survey of protocols & harmonisation initiation	MS1.3: Survey completed (M12)	D1.3 Draft protocol compendium (M12)		•						
	Harmonised protocols & cuisine integration	MS2.3: Guidelines validated (M24)	D2.3 Best practices (M24)				•				
	Cross-lab ring tests	MS3.2: Standards validated (M36)	–					•			
	SME-oriented toolkit	MS4.3: Toolkit finalised (M48)	D4.3 Industry toolkit (M48)								•
WG3 – Data Integration & AI	Platform & tool assessment	MS1.4: Tool survey complete (M12)	D2.2 Prototype database (M18)			•					
	Pilot integration & AI design	MS2.4: Pilot models (M24)	–				•				
	Multi-omics predictive models	MS3.3: Draft personalised framework (M36)	D3.1 Systems integration framework (M36)					•			
	Final open-access database	MS4.4: Database complete (M48)	D4.2 Final database (M48)								•
WG4 – Stakeholder Engagement & Policy	Launch comms tools (website, social media)	MS1.5: Comms strategy live (M6)	D1.2 Comms strategy report (M6)	•							
	Training School & policy dialogue	MS2.5: Workshop complete (M24)	–				•				
	Citizen engagement & training schools	MS3.4: Engagement series launched (M36)	D3.3 Policy briefs (M36)					•			
	Final dissemination conference	MS4.5: Conference held (M48)	D4.4 Final policy briefs (M48)								•

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